

Hamdy Arkoub

Upcoming Postdoc at INL | Computational Nuclear Materials

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EDUCATION

The Pennsylvania State University

01/2022 – 07/2026

- **Ph.D.** in Nuclear Engineering (GPA 3.79/4.00)
Thesis: *“Multiscale Insights into Molten FLiNaK Salt Corrosion Mechanisms in NiCr Alloys”*

Minor: Computational Materials Doctoral Minor

Alexandria University

09/2016 – 06/2021

- **B.Sc.** in the Nuclear and Radiation Engineering (GPA 3.84/4.00)
Thesis: *“Research Reactor Design and its Experimental Facilities Including Thermal Column”*

Ranking: 1st in class 2021, Excellence with Degree of Honor

RESEARCH EXPERIENCES

Idaho National Laboratory (INL)

Starting 08/2026

Computational Mechanics and Materials Department

- **Postdoctoral Associate:** Will begin computational fusion research in August 2026.

The Pennsylvania State University (PSU)

01/2022 – present

Ph.D. Student, Nuclear Engineering

- **Research Assistant:** Conducted research in the Computational Nuclear Materials Group ([CNMG](#)) under Prof. Miaomiao Jin, developing multiscale modeling frameworks for materials behavior in extreme environments using DFT, potential development, MD simulations, and kMC methods.
- **Research Mentorship:** Mentored 6 undergraduates and 1 graduate students in applying Molecular Dynamics simulations to investigate molten salt chemistry and corrosion. Mentorship resulted in tangible outcomes, including co-authorships and Conference Presentations at the ANS.
- **Collaborations:** Granted access to **Idaho National Lab HPC** (till Sep 2027) to support collaborative research with the CMM Department to develop and scale multiscale modeling tools.

Collaborated within Penn State (**PSU Mechanical Engineering, PSU Computational Sciences**) to develop reactive force fields and a first-principles DFT database for property screening and training-set generation.

Summer Intern

- Joined the Computational Mechanics and Materials Group at INL, conducted computational modeling of chemical ordering effects in Ni-based alloys subjected to molten salt using DFT (VASP) and ReaxFF MD (LAMMPS) simulations (manuscript under review).
- Developed and scaled kinetic Monte Carlo model from scratch in C++ using INL HPC systems, implementing parallelization strategies for large-scale simulations to connect atomic-scale simulations with engineering-relevant degradation behavior.

JOURNAL PUBLICATIONS

1. **Arkoub, Hamdy**, Jia-Hong Ke, and Miaomiao Jin. "Atomistic Mechanisms of Stress-Dependent Molten Salt Corrosion in NiCr Alloys." *ACS Omega* 030554 (2026). [\[DOI\]](#)
2. **Arkoub, Hamdy**, Jia-Hong Ke, Kaustubh Bawane, and Miaomiao Jin. "Percolating Corrosion Pathways of Chemically Ordered NiCr Alloys in Molten Salts." *Corrosion Science* 269 113960 (2026). [\[DOI\]](#)
3. **Arkoub, Hamdy**, Daniel Flynn, Adri CT van Duin, and Miaomiao Jin. "Surface Orientation-Dependent Corrosion Behavior of NiCr Alloys in Molten FLiNaK Salt." *ACS Applied Materials & Interfaces* 17(26):38708 (2025). [\[DOI\]](#)
4. **Arkoub, Hamdy**, and Miaomiao Jin. "First-principles investigation of grain boundary effects on fluorine-induced initial corrosion of NiCr alloys." *Computational Materials Science* 255 (2025): 113903. [\[DOI\]](#)
5. **Arkoub, Hamdy**, Swarit Dwivedi, Adri CT van Duin, and Miaomiao Jin. "A reactive force field approach to modeling corrosion of NiCr alloys in molten FLiNaK salts." *Applied Surface Science* (2024): 159627. [\[DOI\]](#)
6. **Arkoub Hamdy**, and Miaomiao Jin. "Impact of chemical short-range order on radiation damage in Fe-Ni-Cr alloys." *Scripta Materialia* 229 (2023): 115373. [\[DOI\]](#)

In Progress:

7. Farshid Reza, **Arkoub, Hamdy**, Alexander Hauck, Adri CT van Duin, Miaomiao Jin. "Electric-field effects on defect migration energetics in GaN." arXiv preprint arXiv:2607.01160 (2026). (Under review: *Physical Review B*) [\[DOI\]](#)
8. Sadia Khan, **Arkoub, Hamdy**, and Miaomiao Jin. "Atomistic mechanism of corrosion-induced grain boundary migration in NiCr alloys in molten FLiNaK." arXiv preprint arXiv:2607.16167 (2026) (Submitted: *npj Materials Degradation*) [\[DOI\]](#)
9. **Arkoub, Hamdy**, Jia-Hong Ke, Nathan Bieberdorf, Miaomiao Jin. "Composition-Dependent Dealloying Pathways in Molten Salt Corrosion" (*Ongoing manuscript*)

PRESENTATIONS

- *“Kinetic Monte Carlo Modeling of Molten Salt Dealloying,”* Hamdy Arkoub, Jia-Hong Ke, Nathan Bieberdorf, Miaomiao Jin, **MS&T 2026 Annual Meeting & Exhibition** (Accepted)
- *“Multiscale Atomistic Modeling of Molten Salt Corrosion in NiCr Alloy,”* Hamdy Arkoub, Daniel Flynn, Swarit Dwivedi, Jia-Hong Ke, Kaustubh Bawane, Adri van Duin, Mia Jin, **PSU College of Engineering Research Symposium (CERS) 2026**
- *“Atomistic Insights into the Roles of Mechanical Strain and Chemical Ordering on Molten FLiNaK Salt Corrosion of NiCr Alloys,”* Hamdy Arkoub, Jia-Hong Ke, Kaustubh Bawane, Mia Jin, **TMS 2026 Annual Meeting & Exhibition**
- *“Multi-Scale Computational Modeling of Molten Salt Corrosion in NiCr Alloy,”* Hamdy Arkoub, Daniel Flynn, Swarit Dwivedi, Jia-Hong Ke, Kaustubh Bawane, Adri van Duin, Mia Jin, **TMS 2026 Annual Meeting & Exhibition**
- *“Revealing the Hidden Drivers of Molten Salt Corrosion,”* Hamdy Arkoub, Daniel Flynn, Swarit Dwivedi, Jia-Hong Ke, Adri van Duin, Mia Jin, **PSU Materials Day 2025**
- *“Orientation-Assisted Corrosion of NiCr Alloys in Molten FLiNaK Salt Using ReaxFF Molecular Dynamics,”* Hamdy Arkoub, Daniel Flynn, Adri van Duin, Mia Jin, **TMS 2025 Annual Meeting & Exhibition**
- *“Atomistic Insights into the Initial Corrosion Behavior of NiCr Alloys in Molten FLiNaK Salt,”* Hamdy Arkoub, Swarit Dwivedi, Adri van Duin, Mia Jin, **MS&T 2024 Annual Meeting & Exhibition**
- *“A Reactive Force Field Approach to Modeling Corrosion of NiCr Alloys in Molten FLiNaK Salts.”* Hamdy Arkoub, Swarit Dwivedi, Adri van Duin, Mia Jin, **MS&T 2023 Annual Meeting & Exhibition**
- *“Investigate the Implications of Stacking Fault Energy on Radiation Damage and Deformation Behaviors of Fe-Ni-Cr Alloys,”* Hamdy Arkoub, Mia Jin, **MS&T 2022 Annual Meeting & Exhibition**
- *A Comprehensive Approach Towards Addressing Issues on High Level Waste Management Plan in the Republic of Korea,”* Jaewon Chung, Hamdy Arkoub, Woei Jer Ng, Naura Qotrunnada, **NEREC 2020 Conference on Nuclear Nonproliferation**

HONORS & AWARDS

Graduate Student, Penn State University

- **Best Poster Award** at the 2025 Penn State Nuclear Engineering Grad Day 09/2025
- **Vice Provost and Dean of Graduate School Student Persistence Scholarship,** 06/2025
awarded annually to a grad student across the entire university.
- **Pitch your GEN IV Research Finalist worldwide,** invited by the Generation IV 05/2025
International Forum to present my research [[YouTube Video](#)].
- **Contributed to NSF-funded CAREER Award** proposal (\$550,000 over 4 years) 06/2024
- **Tau Beta Pi, the engineering honor society recognition,** for being in the top 1/8th 10/2022
of the class of engineering students.

Undergraduate Student, Alexandria University

- **Mohamed Sawan Top Student Prize**, for graduating first in Class 2021 09/2021
- **Aly Mortada Award**, for the highest GPA in junior year, (3.92/4.0) 09/2020
- **Mohamed Elmaghrabi Award**, for the highest GPA in the second year, (3.95/4.0) 10/2019
- **Abdelhamid Elfaham Award**, for the highest GPA in freshman year, (4.0/4.0) 10/2018

MEMBERSHIPS

- **Vice President:** Institute of Nuclear Materials Management (INMM) at Penn State 04/2024 – present
The Penn State Chapter of the Institute of Nuclear Materials Management
- **Graduate Student Member:** TMS Nuclear Materials Committee 03/2025 – present
TMS committee focused on advancing materials for nuclear energy systems.
- **Graduate Student Member:** Material Advantage Student Membership Program 07/2022 – 10/2024
Follows advances in materials science and associated with engineering.

COMPUTER SKILLS

- Programming Languages:**
- Python (6+ years): workflows, HPC automation, ML (PyTorch, scikit-learn)
 - C++(2+ years): parallelization concepts, kinetic Monte Carlo (kMC) modeling
- Density Functional Theory (DFT):**
- VASP and Quantum ESPRESSO (5+ years): interface energetics, PDOS
 - Post-processing: VESTA, VMD (charge density, electron difference maps)
- Molecular Dynamics (MD):**
- LAMMPS (5+ years): large-scale MD, workflows, trajectory analysis, interfaces
 - AMS (3+ years): reactive MD simulations, force-field development
 - Analysis: OVITO, AtomsK, Packmol, custom Python scripts
- Kinetic Monte Carlo (kMC):**
- SPPARKS and custom C++/Python implementations: microstructure evolution
- HPC Systems:**
- Job scheduling & workflows: SLURM, PBS
 - Clusters: Penn State RC (4+ years), Idaho National Laboratory HPC (1+ year)

REFERENCES

Miaomiao Jin, Doctoral Advisor

Assistant Professor, Department of Nuclear Engineering, The Pennsylvania State University,
University Park, PA 16802 | mmjin@psu.edu | 814-865-4863

Adri van Duin, Doctoral Committee Member

Distinguished University Professor of Mechanical Engineering, The Pennsylvania State
University, 240 Research Building East | acv13@psu.edu | 814-863-6277

Jia-Hong Ke, INL Summer Intern Mentor

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